OSAMA SEIF-ELNASER

Mechanical Engineer | HVAC | Firefighting | Plumbing

Abasia, Cairo, Egypt | +20 155 991 7698 | osma.eng.pro.free@gmail.com | LinkedIn Profile

PROFILE

Mechanical Engineer graduating in 2026 from Al-Azhar University (Power Engineering), specialized in MEP design, including HVAC, firefighting, plumbing, and medical gas systems.

Practical project experience across hospital and villa building MEP designs. Very good command of AutoCAD and Revit, with working knowledge of HAP, Duct Sizer, Pipe Sizer, and Elite, and a solid grounding in ASHRAE, SMACNA, NFPA, and IPC standards. Developer of custom Revit add-ins (C#) and HVAC engineering tools.

SKILLS

Design & Modeling: AutoCAD (very good), Revit (very good).
Analysis Tools: HAP, Duct Sizer, Pipe Sizer, Elite (working knowledge).
Programming & Automation: C# (Revit API add-ins), HTML/JavaScript engineering tools.
Standards & Codes: ASHRAE, SMACNA, NFPA, IPC.
Software: Microsoft Office Suite.
Professional Skills: Team collaboration, communication, problem-solving.

EXPERIENCE

MEP Design Trainee — EngX Course
September 2024 – April 2025

- Trained in the design of complete MEP systems, covering HVAC, plumbing, firefighting, and medical gas systems.
- Performed HVAC load calculations and selected indoor/outdoor units; developed duct layouts using Duct Sizer and chilled-water network designs.
- Prepared plumbing and firefighting system schematics, including sizing with Pipe Sizer.
- Applied AutoCAD and Revit for practical design and modeling tasks.
- Ensured design compliance with ASHRAE, SMACNA, NFPA, and IPC standards.

PROJECTS

Hospital MEP Design — EngX Course
Supervised by Eng. Ahmed Shohaib

- Contributed to the MEP design of a hospital building, including HVAC, plumbing, firefighting, and medical gas systems.

Residential Villa MEP Design — EngX Course

- Designed HVAC, plumbing, and firefighting systems for a residential villa.

Office Building MEP Design — EngX Course

- Designed HVAC, plumbing, and firefighting systems for an office building.

Graduation Project — Hospital HVAC Design
Supervised by Dr. Mohamed Younis

- Developed the HVAC system design for a hospital as part of the graduation project.

SELF-DEVELOPED ENGINEERING TOOLS

HAP to Excel
Desktop utility (Windows, .exe)

- Developed a standalone Windows tool that extracts room/zone load data from Carrier HAP (Hourly Analysis Program) output and automatically exports it into organized Excel schedules, cutting down manual re-entry of zone loads during HVAC design.

Variable Primary Flow – Chilled Water Simulator
Interactive web-based simulation tool (HTML)

- Built an interactive simulator for variable-primary-flow chilled-water systems that models zone flow rates and loads, pump VFD speed, and the pump curve versus system curve, while flagging starved zones and pump saturation to support chilled-water pumping and controls design.

Clash Resolver – Revit Add-in
Supports Revit 2020–2027

- Developed a Revit add-in that resolves pipe, duct, and cable tray clashes in two clicks: it splits the element and reconnects it around the obstacle at a user-defined angle, direction, and offset. Works in 3D and all other views.

Print & Line Weights – Revit Add-in
Supports Revit 2022–2027

- Built a Revit add-in that exports sheets to PDF with full control over line weights per category and per Duct/Piping System Type.

EDUCATION

Bachelor of Mechanical Engineering — Power Engineering
Al-Azhar University, Cairo
September 2019 – July 2026
Overall Grade: Good
Graduation Project Grade: Excellent